

ActInf GuestStream #011.1 ~ Gregg Henriques

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hello everybody welcome to the active inference lab today it's october 22nd 2021 and it's actin flap guest stream number 11.1 i'm daniel i'm here with blue and our guest greg so greg thank you so much for being here and please introduce yourself and take it away we'll be monitoring the live chat so if you have any questions during the presentation please feel free to ask them and we'll talk about them at the end but otherwise thanks again for being here and looking forward to this event wonderful hey it's real honor to be here uh i've seen a couple of other presentations they're really fascinating i look forward to sharing some of my version of reality um so let me give you a little bit of orienting coordinates for me um so i am professor uh in the combined integrated clinical and school doctoral program at james madison university uh where i've been since 2003 uh the role i function there is to train individuals to be psychological doctors is the term we use uh the reason we use that is because in psychology in the profession side there are different practice areas uh clinical counseling and school are the dominant practice areas but we want to transcend those into a more general combined integrated perspective that affords us a way to kind of get out of some of the old traditional institutional silos and get a more general integrated picture so i've been there since 2003 my journey in this is i went to undergrad got my major in psychology and was just going to be a behavioral scientist that wanted to be a clinician my sense was the right thing to do was to apply the methods and findings of science to the clinical world i went off got my master's in clinical and community psychology in 1994. and um well why don't i go ahead and start the presentation and then i'll walk you through some of that history and then we'll get to this scenario where ultimately you get this weird cartoon of a garden on behind me and we'll get to there over a period and then we'll start into what we're actually dealing with let me share my screen here so so we're talking about the ontology of mental behavior um and uh just to kind of give it a context um my basic point for defining this talk is then to be in contrast to the current mainstream approach in academic psychology the mainstream approach in academic psychology is really best frame is what i would call methodological behaviorism methodological behaviorism term by jay moore but all it means is basically what is

psychology in the academy right now is that it applies the methods of behavioral science to a thing very vaguely defined as behavior and mental process okay and indeed the methods of science really are what end up defining the term behavior which is the you can see from a science perspective okay uh and then the mental processes are sort of the stuff that's inferred that give rise to the change in behavior at least if you're a mentalist or a cognitivist in traditional psychology as opposed to say a radical behaviorist okay so the science of psychology is really defined by an institution that applies methods of science methods of behavioral science in particular to analyze behavior and then develop models of mental process that afford the explanation now the structure of it is broken at the level of coherence and what i mean by that is it doesn't there's no organizing framework for psychology uh psychology is not defined by and never has been defined by a paradigm meaning a full in a cuneian sense paradigm meaning a full like newton developed a paradigm for classical mechanics that then established psychology has had is a is a field that had lots of different schools of thought many paradigms that tried to manifest itself as sort of the dominant paradigm that didn't happen um there were no dominant paradigms in psychology uh it gave rise to what's called the crisis in psychology that led vygotsky specifies in 1927. so we're talking now over uh almost 100 years ago in fact it was a first identified in 1899 the crisis um and psychology then just punts on the issue it's like what do you what do you mean by the what's the field subject matter and the answer is well different schools of thought we'll frame that in different ways okay the ontology of mental behavior is the argument that actually we can do better than that um we don't have to define the field based on epistemology i.e the methods of science but we actually can define the field based on a scientific ontology that clarifies what is meant by mental behavior and in so doing it affords a coherent solution to the problem of psychology so the problem of psychology is the inability of the field to cohere around a particular conception of its subject matter so that's the basic structure um i'm like i said i'm at james madison combined integrated so what's the basic gist just to get a lay of the landscape from where i'm coming from and i'm you know i'm an odd bird so we should be very clear that i i've been doing a weird version of psychology now for uh for a long time i started to say this i entered into the field sort of in a conventional sense i got trained in the normal methods of behavioral science and analysis of variants and and all that good stuff and it's like oh well clearly we want to employ a standard empirical method to delineate you know what the sort of the truth is or the most accurate models um and so uh but then i encountered this thing called the problem of psychotherapy uh what's that we actually get into the room

with people and that's that's a trick um and then you're a real person in a real situation in the idiosyncratic processes by which you have to make active inferences about what is operating in the moment okay um means that the generalized aggregate variable analysis that the rest of the field does like oh here's a problem it's called major depressive disorder here's a specified intervention called cognitive therapy and now you're to imply this cognitive therapy intervention with this person um sorry but what i learned and i didn't know it at the time but what i learned was the level of abstraction at the variable aggregate level relative to the actual process by which you engage in feedback in the room in a participatory real relationship of course i have language for this now that i didn't have them but the bottom line was god you know uh what i actually saw in the field was that the best of the best depending regardless of the school of thought the best of the past did the best surprisingly enough and lousy players from various all the different schools did lousy and most people did average and the schools of thought really don't generate um systematic difference at least the bona fide schools of thought so essentially what you get in psychotherapy is what's called the dodo bird finding uh i'm president-elect of the society for the exploration of psychotherapy integration which is a field that has long standing questions in relationship to the relationship between the schools of thought um and uh and and i got then introduced in 1994 this is basically to the field through a potentially integrated perspective for me that was fascinating um it turns out that my orientation is actually what's called a big picture coherentist orientation meaning that i actually want to zoom way back organize pieces in a particular way so that i can create a coherent frame of reference and then zoom back in with the coherent frame of reference that's the way my mind operates so i had zoomed out and then realized there was actually no coherence um at the level of psychotherapy and people had known this for a long time and then this whole field that emerged in the mid 1980s this was like you know 15 years 10 years down the road from that emergent of that field with the psychotherapy integration problem and then i was fascinated by it and i did something that's actually no one's ever done which is kind of interesting um i looked at medicine and this is just all intuitive i looked at medicine um and i said you know you don't have all these different schools of thought in medicine it's not like you have you know the the cardiologists are not battling with the endocrinologist um versus uh you know the people that emphasize uh the gut you know they all are like hey they're different systems of you know biological adaptation that function together in a hole and think we have a thing called human biology human biology is the science that organizes the various domains um and that affords a descriptive explanatory framework that then should inform the

practice of medicine so if we jump up a level it seems reasonable to me or it seemed intuitively reasonable to me that there should be a human science of human psychology that should organize the descriptive explanatory network of understanding okay and then apply that to the field all right so now notice what i had done i got socialized into the empirical methodological behavioral science approach of psychology dropped into the problem of psychotherapy backed up and was like oh wait a minute what i really need as a practitioner is i need a holistic conception of human psychology that if ford's clarity about what the hell the field is actually talking about sorry i'll probably drop an f-bomb here or there if that's okay but anyway what i basically realized then here is i shifted perspective and i said well what is psychology and then i realized oh my god nobody knows what the psychology is there's no word that everyone's got different reference points for this fundamental word okay and that's the problem of psychology there's a dramatic set of reference points um that and so i then uh took a couple of left turns that i can tell you about in 1996 and then a big one in 1997 and all of a sudden i actually developed a vision um not like some transcendent vision although it has these elements i developed a picture an organizing picture really a new map of big history as it were that afforded me a new way to address this problem of what psychology is okay and that ultimately gives rise to a clear ontology of the mental all right we'll specify what i mean by that and then by affording clarity about what the scientific conception of the mental is i could then uh organize my work and return to the original problem and develop a framework for psychotherapy uh and so that's what the project is about now my journey over this time period basically is the first thing is hey here's a problem and then i need this and then i built this thing called the tree of knowledge system the tree of knowledge system is this natural scientific ontology that maps the world in a particular way that affords a science of a natural science view of what the mental is um i worked on that and said hey it organizes the field um i did a you know i can tell about my publication history etc and then ultimately that evolved in what's called the unified theory of psychology the unified theory of psychology specifies the tree of knowledge and then a couple of other key ideas that then provide a conceptual architecture for assimilating and integrating the schools of thought and major empirical findings into a coherent whole i'll say again it provides a conceptual the major major schools of thought and major empirical findings to afford a coherent poll of the picture and basically then create a new unified theory of i then shifted that i did that that published that in 2011. um i then returned to the original problem um and from 2011 2016 or so worked out then given this framework what would a unified approach to psychotherapy be and i then delineated four key

ideas that could help organize the field and generate a unified approach to psychotherapy at least with individual adults okay children are different blah blah blah i was honed in on sort of standard individual psychotherapy especially for neurotic adults which are the vast majority of people come in for psychotherapy or neurotic adults okay so this forwards then a one-on-one a way to organize the key insights and then and so then you would have a unified theory of psychology and a unified approach well then in 2016 as i was delineating that i took another weird turn all right um and there are a lot of reasons as to why uh but um the short story is as i was at a conference uh called cultivating uh the globally sustainable self um about transformative education and the massive global problems that we face in relation um and it was in the context of that conference um i had created this model the unified theory and unified approach it was called a tua okay that's it just dropped into the language you know how things would get more and more efficient you know it's ut and ua and you tattoo uh somebody's like what the hell's your two lines like how's this video blah blah blah so i'm at this conference and they're um uh doing this talking fundamentally about how to do this thing called cultivate a globally sustainable self meaning how do we transform our education for children in the world that we're in given all the crises and challenges that we face what does that actually look like it was fascinating conference and at the end of it there was a panel discussion okay and panel discussion actually got all the people they invited there and they came up and they actually all fumbled over the question as far as i was concerned like they were brilliant people they did this they couldn't actually there was there's a summary opportunity i left the conference and i was walking away and i was talking to somebody i was like you know what we should do we should plant it to a seeds and grow it to a trees um which sort of fell out of me as a light bulb moment and then turned into the thing behind me which is this garden so all of a sudden i basically then transformed this build this construction that i had with all these different sort of like a cathedral that got transformed into a garden i actually then got called into like spiritually artistically to build this garden so that now this knowledge system then could be represented symbolically in a garden and that's what spine is a it's a it's a layered knowledge system so like for example well i'll just so that that's that's and then you know i got pulled more into that and then that shifted me further and further into the orbit of weirdness all right but anyway that's basically what i realized fundamentally is why i got pulled into this level of abstraction um was something that i sort of got clear on as i've been writing my current book and that is a real understanding fundamentally of what's going on here in relationship to kind of our knowledge system so we have a foundational problem in our

knowledge and what i call in this foundational problem is called the enlightenment gap what the hell is that the enlightenment gap is the point that when science emerged okay um in from galileo and newton coming off of the scholastics okay when we built physical science and the epistemology and ontology that emerged with physical science um there was never a ability to take what we learned from the physical sciences and create a holistic synthetic philosophy can't tried hegel tried etc but what you basically get in the enlightenment is a gap a failure to get the proper relationships first on the relationship between matter and mind okay some base mind body problem what the hell is that how do you actually solve that and then the other thing that emerges as a secondary joint problem is what is actual scientific knowledge relative to social knowledge and this problem emerges we see in the 20th century with the emergence of the philosophical post-modern critique of modernity which basically says hey modernity made too many abstract pure reason progress claims that they were actually that you could find genuine truth that's not contextualized in a socio-historical justification context that's the post-modern critique and actually you can't really make knowledge claims that are transcendent they're all imminently contextualized in the socio-historical context and lo and behold look at the way the knowledge systems emerge they get they privilege cis-gendered white colonial men who go around and create oh yes this is true reason and behold it's actually it's normative for us and we get to distribute our power structures around the world and so you get a philosophical and a social justice post-modern critique about what science is relative to social knowledge that never actually gets resolved okay that's why we have post-modern versus modern onto epistemological debates about the nature of knowledge and so the enlightenment gap basically is a description of the current state of affairs where we have we're lacking a philosophical framework that affords us clarity about matter and mind on the one hand and science and social knowledge on the other so that's the gist of the location about where we are um i built this thing called the unified theory of knowledge okay what is unified unified is taken directly from the book consilience the unity of knowledge e.o wilson wrote a book called consilience in 1998 called the consultancy unity of knowledge that's what unified means it means conciliating there's a consilient coherent big picture view that actually gets the pieces of the puzzle correct so we can actually understand knowing okay and knowing how we know and knowing what we mean when we know things physically and what we know things biologically what we actually mean psychologically human psychologically into the social sciences how we know scientifically relative to how do we know subjectively what do we know about ethics etc there's actually a coherent synthetic philosophy that is available that

frames these problems that wasn't achievable before and so here's the garden okay uh and what it has here is this so for example right here this little thing's called the stepping stone what is that s-t-e-p-p stands for the standard theory of elementary particle physics and the stepping stone then grounds you in physics in a particular way then this first branch is the tree of knowledge system the base of which is the energy information quantum field structures that live underneath matter and go back to big bang and then each one of these is a layered system of information processing called life mind and culture and it is that arrangement of energy information gives rise to a classic material field and then life emerges and then mind and i'll explain what i mean by mind here emerges at the animal mental behavioral level and then culture emerges so that we become go from primates to people as we talk to each other and build what are called systems of justification okay so that just gives you a flavor for it there's other things that are going on here that i won't necessarily get into but it basically says there's actually a knowledge system here the left side is the unified theory you can see actually there's a little u here and a t and these four ideas make up what's called the unified theory there's a u and an a which stands for the unified approach okay this right here is a symbol of mindfulness on the right side of the branch is the way in which we relate to people in healthy psychological ways it's actually called calm mo it's an integrative approach to psychological mindfulness it's the processes by which we can hold each other in ourselves and deal with difficult socio-emotional relations that afford productive virtuous looping as opposed to what often happens people as they get trapped in negative cycles attack each other and then get in vicious destructive looping that's just one example so the tree then has eight key ideas four of which make up the unified approach this is by the way the tree of life the tree of knowledge is the first branch tree of life has four here four here and then as this symbol in the middle which is actually called the metaphysical empirical flower metaphysical empirical flower creates a dialectic between metaphysics and empiricism and that was missing in the structure of the enlightenment which we need there's another thing down here called the coin but i'll pause here just so there's plenty of um structure there let me make a few more comments and then actually i'll pause after this so ultimately what the unified theory does is it affords a new opportunity to think about the enlightenment gap and i are you resolve it um and what it does is it sets the stage for a unified theory of and frames a unified approach to psychotherapy okay um so that's basically then we can get into what does all this mean in relationship to some of the deep theory but i wanted to give folks an opportunity for the basic gist of where i'm coming from thank you for the summary cool work and symbolic images so anyone watching

live can definitely write questions but that's a great place to stop we got the personal and the multi-level narrative coming together with a little bit of a challenge and a response which we'll then learn more about so maybe we can just go blue if you have any comments otherwise we can go into the next section um i don't know if we want to go through some of my questions now i mean i have some that relate to the uni unifor universal theory of knowledge um so if we want to maybe pull up that slide or if uh greg is still sharing slides we don't have to i cannot just read it off this slide so the first perspective yeah and it's on the screen as well so here's um uh one just summary quote first the the ut okay differentiates the mental behavior of animals from the behavior of other living creatures both metaphysically and metatheoretically which you mentioned and second it differentiates the mental behavior of human persons from the mental behavior of other animals both metaphysically and metatheoretically okay so similar but a little bit different there so what are the distinctions that are critical for psychology what distinctions does your work make how do we know those are the right ones how did you carve ut okay at the jigsaw joints beautiful okay so yeah i love that that's a wonderful set of questions and right gets right to the heart of some of the key issues okay so let's talk about the so the the tree of knowledge divides is a descriptive metaphysical system that's a um the picture is and it shows matter life mind and culture okay so basically coming off big bang so you start at what's called an energy information singularity all right that's how you go back to big bang 13.8 billion years ago et cetera you for you run the tape back and everything collapses into quote-unquote a point or a singularity because this is just general cosmology all right i'm getting to the question but i just give this background okay so according to modern cosmology the universe essentially begins in a state of pure energy information that's at least what i refer to it as we can talk about the term and then out of that comes matter okay uh and basically what you get is you get all the four fundamental forces in nature that's electromagnetism gravitation strong weak differentiating to create fields and then you get a coalescence a cooling and freezing and coalescence of energy into chunks of matter in the form of particles atoms up into them chemicals in certain unique contexts and then across scale that builds things like stars and galaxies and mega galaxy clusters okay so this is then our physical universe all right and notice what i've done and the reason i had to say this is because one of the first things we have to note from classical mechanics is that modern physicist physics teaches us that there's a thing beneath the classical material world called quantum field theory and the energy information singularity of big bang the reason that's really important because we have to understand the fundamental substance ontology of the world

which is energy information fields not billiard balls okay so they were working in the initial sort of ontology is billiard and it's like no that's actually not the ground of the universe ground of the universe is sort of like an energy information super force field theory all right which by the way for mental implica the implications to understand the way mental structures work it's it's important that some people like bernardo castro drop it all the way around call that big mind i don't agree with that but fundamentally we do want to understand that okay then we come and we jump up we don't know how this exactly happens but about 4 or 3.8 billion years ago we see the jump in the world from chemicals complicated chemical protein lipid structures that someone get jumped together organized and we get life okay so the miracle of life all right so the miracle of life then creates the jump from quasi-organizing systems that then get into fundamentally slooping into self-organized what deacon terence deacon calls an intentional phenomena that operates across abstentia okay so intentional phenomena meaning that basically there is some descriptively not explanatory but descriptively these cells behave with a goddamn purpose they're doing okay and it somehow is that they're imagining that they are black and then they move toward the lack like they're hungry and they and they don't want to die and they move around okay incomplete nature is terence deacon's analysis of this jump well this is really important to understand the emergence of complex adaptive dynamic systems because that's what life is it's a complex adaptive dynamic system that has embedded in it energy information systems that afford the capacity to pull i'll use a slightly different color first and free energy from the uh environment to do work to maintain its complex organization so the survivor reproduces and the reason i have to tell you all of that is because the ground at what we need to understand the bi dynamic biological ground out of which then minded animals are going to emerge so so that we get a natural selection operating on genetic combinations and what happens according to unified theory is actually you get an orthogonal dimension of activity orthogonal to the real in some ways what do i mean by that it's the subjective information processing the epistemic information processing of a cell okay it's like the cell is actually making judgments and quasi even decisions as it engages in some kind of primitive act of inference um and so then that becomes the ground and then basically you get bacteria that are doing that for a long time and other kinds of animal kingdom i mean uh you know algae and other kinds of protists kind of kingdom stuff and then you get multicellular plants okay um that afford the opportunity for what big historians call big life so you get cellular plants and you're also going to get fungi um mycologists study things like they do really weird stuff uh fungi um and then you get so you're gonna get these active multicellular

entities that are still though fundamentally organized and grounded and change their environment through their growth potentials within the cell so even things like mushrooms what mushrooms don't do is they don't get up and walk around all right um but what they do is they send out growth signals into particular things and then they grow in really interesting ways but they are confined by their boundaries but what happens in the in the animal kingdom as a shoot-off in relationship to this so that this system now becomes as a multi-organizing system becomes mobile all right so 700 million years ago you're going to get then a transition into basically jellyfish-like creatures okay that have distributed neural networks that have some degree of capacity for coordinating their behavior all right through the distributed neural net that's the first step towards a minded animal it's not there yet okay you then get bilateral planning with an in an emerging integration between the sensory motor structures okay like in a planarian worm now a planaria worm's got a centralized controlled center okay that's then distributing and it's got bilateral movement to coordinate its movement in a particular way now you're very close with a planaria worm to what i would call a minded animal the final thing is that you then got the cambrian explosion 550 520 million years ago and in this short period of time basically you get a much greater complexification of the nervous system you get a centralized nervous system and you get a complex active body plan that complex active body plan then regulates a sensory motor looping okay that affords the animal to get up and move around all right now that's a game changer it moves the entire structure as a sensory motor loop okay and it is that division that joint point okay between life living organisms and minded animals that the tree of knowledge points to and says hey that cambrian explosion thing is when minded animals blew out of living organisms there's a joint point thing that's as important to understand in our big picture as the joint point between non-inanimate matter and matter and it follows a very similar pattern because what you see like with cells what did you get you get an rna dna information processing closure around a lipid with protein generation that then creates a communication network between other cells well the nervous system is very much akin to the rna dna it's a centralized control system and what it does is it yokes all the cells together in a coordinated way so the animal gets to move as a whole unit so the sensory motor looping of animal movement then becomes a fundamental another dimension of information processing communication and we get animal animal movement at working with each other in a communication network you get the base of the mind dimension okay so in the unified theory that's and you can and then it's defined behaviorally and descriptively so it's just a complex active bodies moving in relationship to each other with functional outcomes that's

what it is so when you go and you walk your dog your dog is exhibiting this the insects are exhibiting this and it's it's called mindedness or mind it's the functional animal the behavior as a whole mediated by the nervous system okay it produces a reliable understandable effect the base of this looping i see as basic you know predictive processing active inference essentially okay there's a fundamental iterative process that is regulating information energy in particular structures that theory the meta theoretical structure i bring to bear is called behavioral investment theory okay behavioral investment theory is a meta perspective that takes together what's called energy economics so you have to solve the problem of uh entropy and energy expenditure so first principles energy economics second is evolution the natural by natural selection and other any other things we have to add to that but that's the general gist the third principle is behavioral genetics which means the genetic the gene system is going to bias the system in a particular way the fourth principle is neuro-computational control so neural computational control then the processes by which we engage in neural computational control then is analyzed deeply by an active inference predictive processing the fifth principle is learning and of course then that's how a system upgrades in relationship to experience and the sixth principle is life history development meaning depending on where the animal is in its life history is it young is it adolescent is it late life et cetera it's reproductive and survival strategies are going to change based on its life history development so those are the six principles of behavioral investment and what they do then is they yoke together neuroscience and evolutionary theory and computational cognitive and by the way behavioral science from the vantage point of a radical behavioral position like a skinner this is actually i come through this skinner by the way which is weird okay so so skinner actually what skinner does is he just sits on the outside he's he basically said oh it's black box it's physiology black box but what i will track is i will track at a very fine-grained analysis consequences the animals doing something has got consequences and it is the consequences that pull the animal in its behavioral investment and i can navigate the consequences as an instrument as an engineer he thinks like an engineer he's not thinking like a theorist he's really an engineer you know you want to be a poet okay but it basically fails as a poet and he's like well what really but you know really we can do a psychodynamic interpretation what happens to him but actually then he starts just focusing on how to control the consequences to pull investment from an animal and so that's an exterior behavioral position and that's radical behaviorism which by the way is totally different than mythological behavior and it's a very you know five percent psychologists go this week applied behavior analysis but it's really key because we want to be able to bring the

fact of the matter is the dynamic control consequential system from the outside behavioral outside is going to connect to the after active inference so somebody asks there's questions like hey can we connect enactment and an activism and representationalism with a behavioral selection structure yeah yeah you can okay so anyway that's behavioral investment theory and what that does is it takes you from basic the cambrian explosion and crabs all the way up although everything's changing but you can go all the way up to primates following this line and that's the dimension of mindedness we can talk about the stack but behavioral investment theory tracks that stacking the evolution of mind brain behavior in the animal kingdom it's called mind as a capital m as really an introduction of a particular term and that's the first joint point and so what the descriptive metaphysics and ontology is oh here's the set of mental behavior the meta theory's behavioral investment theory okay and that goes then from organisms into animal mental behavior and it's a joint point that people fail to see too often so i'll stop there and say this is then the ground of mental behavior is we've set up and then we'll go all the way up through the animal kingdom and then we'll jump into the human primate to become a person and what that does and then finally we actually got to come back and actually talk about the psyche but that's a whole nother the whole another step so so just to clarify here like you're you're presenting two things as central and kind of required for mind with the capital m and those are mobility and a central nervous system is that correct right it's really it's a complex active body um with a central nervous system that ties up a motor system in a sensory system together there's interesting an analysis of evolution about how you get a sensory system and a motor system how do they actually evolve when they come together so it's when you have an exemplar of a minded animal is a complex centralized nervous system sent active body that produces a functional effect on the animal environment relation well so how do you rectify like things like people who are paralyzed are they then mindless because they're not mobile oh of course not so then we need we'll dialogue exactly so so now we're gonna then dialogue about the different domains of mind okay so ultimately we'll take somebody like you know um stephen hawking okay so seaman hawking had an entire shell of existence right i put through the mediation of the sensory motor control system on the behavioral exam to lou gehrig's disease unplugs that you know what a nightmare right and then you get what he then experiences what's called mind he still has in my language we can get to it mind one is inside the organism the neurocomputational layering and the modeling inside the organism is called mind one and that's the neural computational structure okay if you mind one a is the inside mind one b is the mediated by the sensory motor looping

definitely you then the entire definition in the large scale of what the exemplar is is the sensory motor loop but obviously then what happens is what's called mind1a is the neurocognitive infrastructure within the nervous system okay versus mind1b which is the actual observable from the exterior mediated by the muscle movements so what happens is with people that would get this is mind1b gets x okay but mind one a the architecture's still there now if we now jump up to humans they have what's called mind two like other primates and i will we can argue how far it goes mind two is called subjective conscious experience of being this first for the hard problem of consciousness the first person experience of being mind one gives rise to then we can dialogue about this mind two mind two is available from the inside subject of conscious experience and then mind three is the justifying mind which other animals don't have but humans do okay and then we created informational interface with stephen hawking so that he kept after his entire system got shut down instead of locking him in we were able to create a verbal interface so we can maintain contact with his brilliant mind and he could tell us about actually i don't think the information is lost in black holes and i can actually even do the analysis of it okay but if we didn't have that interface he would have been lost but we would have known that yes he would have had mind one a the inside inferential architecture mine two the first person experience of being and mine three a which is the internal narrator but he would have lost mine 3b which is what i'm doing right now talking to you publicly but thankfully we maintained his information interface and uh maintain contact with his great mind thank you greg so um i just wanted to kind of recap two points and just maybe show a little point of contact with what you mentioned in some of the big themes in active inference so first you mentioned how there was these abstractions and mean field approximations in the clinical field and so that didn't that was like speaking to a disconnect between theory and application and practice and making impact on the systems that people care about and then also to the more deeper physical question about mean field approximations and whether you can study like the trajectory of one particle versus just take the time average the space average we're not going to go into the whole ergodicity non-equilibrium steady state definition but those are key terms for active inference so they come into play with that difference like i i'm i am i don't have mastery around the on the active inference meaning of a lot of those terms so i need to get i need to be brought up to speed on some of that but it's exactly that if someone says well you know the machine in the mean field it works but what if one in 100 fail catastrophically then is that an approach that works for example so we've seen that come up in some machine learning context so that was kind of a cool um showing how the idea of the part

particular trajectory versus the mean field and how that plays out in the way that our knowledge and application systems health care systems are constructed which is a niche modification through us and then um so then the second one was like that combination of appeal to general frameworks universals and the historical trajectory of the horizon of life on earth and of course our developmental trajectory as psychological individuals so then it's like well that looks goal-oriented a lot i thought goal orientation had to be purged well i mean the first easy answer is just that was a period of disinformation and that that stance is simply incorrect so that's one way to simplify the debate but then the other maybe longer answer you said it's energy information fields not billiards so we don't appeal to anything when we see diffusion and ink diffusing now that's kind of slam dunk but are there diffusion processes for metabolism strategy narrative communication so why don't we need anything more to just allow those flows to also happen and so in active we talk a lot about those types of models as well so it's totally a lot of cool areas nice yup first section yeah no okay so and the tree of knowledge is both continuous so it allows for full deflation on the one hand or reductive continuity uh into energy information on the one hand and then if you look at it it's actually discontinuous dramatically so in its funnels matter like mind and culture and so there's a profound discontinuity as well and it is the process by if we really want to get at this it's the process by understanding this very complicated dynamic of continuity and discontinuity and the the unified theory affords a particular way to understand the discontinuity of matter out of energy life out of matter mind out of life and culture uh out of mind um and so yeah but but the the reason when i heard carl you know especially free energy stuff although i still am trying to wrap my head around all the terminology and i haven't devoted myself to really uncovering it okay but the the tree of knowledge basically is a depiction of negan tropic curvature okay uh and then the question is how does a system pull how to via tracking forms it actually functions to track forms and make predictions and it's got to create anticipatory structures that both navigate the dialectic of stability and plasticity to pull free energy into the structure and i use that term more in an eric liaison way he's a cosmic evolutionary biologist i mean a cosmologist and talks about the evolution of mega entropy through a free energy principle um and because i learned that first and then i was i've been entangled some with some of the confusion on at least classic uh call fristens free energy but anyway it's it's very close i mean i just i'm sure it's just a technical uh set of terms i need to download to be clear about well we and many others including anyone who listens to this are part of the sense making and we're working on a participatory ontology there's a lot more to say but let me just

give one free energy answer and then maybe let's dive back to a presentation of yours yep so um to return to this idea of energy and information fields being primary not so in a world where matter energy is heterogeneously distributed some is in stars some is in interstellar space or whatever systems that resist dissipation what we use the blanket terminology to discuss markov blanket systems that reduce dissipation and have a non-death equilibrium stationary state those systems are at least acting as if they're trying which in complex symbolic ecosystems means anticipatory predictive behavior anticipatory predictive signaling and therefore it's helpful to have an action loop model perception cognition action niche impact it's helpful to have that type of model which is in the pre-active inference work of Karl Friston and others also central the dynamical equations and stochastic um simulations and things like that so there's a lot to say but that we all hope to be resolving that uncertainty in the country great i'll pop into where i'm coming from and this real fast i'll share screen go to different awesome there's a um i think i need to go here hold on one second uh because i wanted to pull this up and did i maybe i closed it oh well um yeah i'll take a second here one second let me just look at this i thought for sure i headed up reset your priors yep this is more complicated than normal let me see if i got it right here okay if i go here and then one other short thought while you're oh unless you have okay go for it i'm gonna just close that down one second i'm sorry oh yeah take your time just one other short thought blue do you want to say anything first okay um Tim Bergens' questions part of a pluralistic approach towards understanding includes mechanistic and historical answers and so the way that you situated this in the sort of ecological critique and the socio-ecological critique the relativist and not just like nihilist like relativist things are relational its interactions all the way down rather than particles all the way down so that and the dialectical tradition as well so many interesting threads so appreciate how you concisely discuss that great um yeah and i'm happy to pick that up sorry for my clumsiness here but this is so here we go in terms of the basic structure uh that you will recognize um so here's the i came at this for what's called perceptual control theory i don't know if you know not too many it's underdeveloped relative to um at least in my fields clinical psych really doesn't pay any attention to it well William T. Powers developed perceptual control theory it comes off the old cybernetic tradition and all a lot of this will feel uh similar but anyway behavioral investment theory then evolves in from a perceptual control theory model which i see is very very much in line and essentially what you're doing here this is the basic schematic i don't get in i'm not well versed in mathematics my son's a math major he teases me all the time i can't even god damn you have differential equations which is to my shame but

whatever it is part of my reality um the bottom line is is here's the basic schematic okay so you get saline control variables you get an input transduction function and you're going to create a model perception that's going to create a valence valuing structure that's an energizing action in return so this basic loop is the loop that i've been operating off of and basically it's an active inference uh control loop uh in you know not without getting under the hood of the neural computational specifics about how things are set in terms of its sensitivity um but you can follow this uh at a more sort of psychologically higher level of organization um so that's what i just wanted to show is that it's the the ground of behavioral investment theory is this perception action calibrating loop structure that functions and an example in perceptual control theory is you create an image of an ideal egg and then you run through a bunch of sub goals in relationship to reducing discrepancy towards that you having emotional reactions in relationship to whether or not you're achieving the efficiency of those goals in expected time frames as long as you're not getting surprised and in the constraints and moving in the right direction you're getting jolted with positive affect and continuing the investment if all of a sudden you know you break the egg the wrong way you overcook it there's no eggs in the fridge they can interrupt the structure you get negative activation of affect and then reboots a possible new behavioral investment path um so anyway just in terms of sort of the basic architecture of uh that structure that's actually a perfect short section so thanks for making that clear let's get the active loop up mention uh two points of contact with active and the loop that you described and then blue um so the first one is that in active the preferences are over sensory so that's very much and i don't know which um if we had a unified knowledge management system theory of knowledge then we could search through which discussions we discussed it but perceptual control theory did come up because preferences are over sensory outcomes so they're not over hidden states of the world which are inferred using latent bayesian models but preferences are over sensory outcomes so that relates very well to the embodied approach and then another just key point of contact which um in the bayesian space it might not feel like this is a major thing but relative to the development of ideas it is and that's related to the experience is perception and the experience of perception is generative so like a neil seth's being you new book summarizing it as well as many other approaches like we're experiencing the generative half not the sensory incoming half which is the total signal processing info out there model of the world name in yeah that's that's all part of so i have uh um in terms of my own i've now comple i don't know if you guys know john vervicky's work i don't know if his work has come up at all uh so me and john have completely unsynced up we

found each other 18 months ago or whatever and he built um you know cognitive science and philosophy he built what's called recursive relevance realization as a core structure that builds right into predictive processing you know and then the uh what level that like you then that becomes really refined in terms of the difference and i certainly get john here if he's got time he's really busy but the bottom line is that that john's recursive relevance realization model meta theory of cognition and his taxonomy of knowing in the 40s got four peers of knowing are now completely synced up with the unified you talk okay we did three different video series one on consciousness one on the self and one on human transformation the last with uh zach stein who's a developmental metapsychologist educational metapsychologist on expert transformation so anyway these systems now are affording all sorts of productive interplay um in relation so they now and it's a hand in glove relation i mean he comes out of cognitive science and activism and philosophy i'm coming out of psychotherapy and clinical psychology into the science of psychology and really from a behavioral and evolutionary perspective as i got dropped in neo behaviorals and really it's where my training is roots so anyway then you get so you get an activism there you know and his stuff connects so it's a real there's a real matrix of connection that's actually operating blue nice that's like a perfect flow into um the question that i had related to both active inference and in activism so one of the like fundamental things that active inference does it kind of um unifies this inactive embodied approach as well as representationalist approaches like where you know there's mental mental representation centered around mental representation um through this like action perception loop which is essentially behavior i mean in some way and so i just was wondering what has active inference been fitted to like this unification of mentalist and structuralist and behaviorist approaches or do you think that that would be an appropriate way to leverage active inference or how else could it maybe be used in your work well yeah no that's exactly um so i i totally see that i totally see it as you know the representational and an activism in fact john and i i'm fine with representation john likes modeling as a better term rather than than representation a little uh he wonders a little bit about some of the history of that term you know it's got it's one of the classic terms that means all these different things but at a generic level i'm totally fine with term representation uh and indeed i'm sort of a neo behavior is coming off an old cognitive simplistic cybernetic model which is basically all representation you've got a mentalist model of representation and then you're mapping the world cognitive maps from you know uh tolman all the way back to that so for me that bridging is very clear and necessary and brilliant and beautiful that an act of inference does um and it's just one of the network

possibilities of different kinds of schools of thought that are afforded and the unified theory basically what i focus on essentially is the class of cognitive approaches weakly defined and then really that the thorn in my side was skinner i was like hello he gets skinner over here okay and at first i just axed him i was like he and you know he totally negates the agent and information processing from the within in a totally unacceptable way and therefore his philosophy of ryan is wrong after all trump's killed him anyway um so that was my justification then i actually learned what skinner was trying to do i agree with everything i just said by the way but it was also the case that's that skinner was also on to an angle um that red that ultimately sits in psychology as the fundamental mental behavioral kind of schism and ultimately the articulation of mental behavior as the ontology of is the way in which you conceptually resolve the underlying disputes in that structure all right so my basic point is yeah 100 we can actually weave all of these schools together now and afford a link a sinking a synergistic linking with coherence that was never afforded before and obviously active inference doing that with representationalism and activism is great it's a piece of the puzzle we can now sync that now that i'm synced in with john this thing all loops back around i'm telling you this thing's about ready to blow we have too many people in uh the active inference field that are also modelers representationalism is a useful term as opposed to modeling because it's uh you know it's modeling is like what i do to of representation or i don't have a i live in my representations and my models and they hang out fine together a lot of a lot of things i wrote down just wanted to give one point of difference between the perceptual control theory and active inference and maybe some of these other approaches because i understand that those are commonly used or at least known so if someone came to the you know department of xyz and said you all know perceptual control theory perhaps um missing in that loop was the whole machine learning type work of policy selection it just went from basically like this to inference and then emotion but it's not just about how you felt when you looked in the fridge it's about the multi-joint mostly sub-personal sub-experiential components and that's what actually provides the kind of broad shoulder that connects cognition across life and maybe even non-life is um gives a continuity to grounded in reduction of uncertainty conditioned on policy which is what active formalisms are about and there's a lot of awesome diagrams and pros and theory that's built up in the qualitative fields and now we have new tools and new approaches love it there's some questions in the chat or you can perhaps present any more slides well i'll at least just tell the rest of the story in terms of what uh what we're so now we've gotten to minded animals okay complex active bodies running around after the

cambrian explosion um and they're engaged in the sensory motor looping battling with each other for territory control etc okay then we can i'll be happy to stack that i basically say that we started off in the planaria where these reactive animals okay and then you get into real learning animals um and and i believe that the in the cambrian which are these dynamic feedback structures that are a lot more sophisticated and they engage in operant shaping a good uh book which is i think right here uh that i recommend uh the evolution of the sensitive soul um by uh ginsburg and jablanca it basically explores the evolution of that nervous system and what they call basically a flexible learning operant system that comes out of the cambrian explosion it's on whether or not consciousness gets a grip in that term meaning when i say consciousness i mean mind two that's the phenomenal that's the ex um nagel's experience of what it is be like um and basically from my vantage point you get pleasure sensory pleasure pain you know jolts as the first kind of what john and i call valence qualia okay probably are the first sets of mine too and and i'm an early mind to her which says yeah maybe even down to insects okay um but that's a huge debate uh we then climb up the structure of the brain and then get into a much more extended inner life and i certainly think by the time we have crows and other mammals and things like that you get a rich mind to inter perceptual field um and then they attract that into primates and you get you know you can defer you get real relational dynamics this was a question that came about what about relation and would talk about that um but you get relational dynamics with parenting first and then you get a parent in a mammal system you get a parenting attachment system where you're doing self-other modeling and our care around another in a long-term relation and the attachment system comes online you then get social animals you get competition and cooperation and you get this really rich social primate field um and then i do tomasello's work on the emergence of a real inter implicit inter-subjective we space field where there's actually this capacity for massive flexible modeling of the other and mind reading shared attention intention space this is about 500 000 million years ago when we're you know pre-fully verbal um you know prominence running around coordinating to kill mammoths and like that okay um and so that it's and it's qualitative we're a unique great ape at that time uh because of our implicit group shared attention all of a sudden we can actually like work together to create a coordinated shared community felt sense of what we're doing together this is pre-linguistic and then but what it does is it creates a shared mind space for symbolic tagging okay that's what happens first and music is probably happening etc although that's a good question about when and so now the unified theory picks this up uh at this stage um so there's all the other great work that other theorists are doing and

then it says well you get symbolic tagging okay antelope there and you're coordinating the group and that's all very intense requires a lot of definitely high cognitive stuff that's probably pressing the evolution of the brain in a particular way the argument basically is 200 000 years ago probably and we'll see if this gets pushed back but certainly by 50 000 years ago and this is the window here it's a fundamental shift in symbolic communication from a broken symbolic system into a symbolic syntactical system okay they're basically a chomsky and language acquisition device at least in the weak version of that all right that then affords the capacity to place nouns verbs and subjects and objects in a propositional statement this is a game changer the proposition according to unified theory why because actually a proposition then is a claim it's a propositional claim about a truth statement on the one hand so the antelope are over there the antelope are over there then organizes a particular truth crane that opens up the possibility for counterfactual negative space in a radically different way no they're not okay those are gazelle not antelope or and what it does is it opens up the possibility no we should go hunt rabbits so propositional claims and the counterfactual structures that they then once they take a positive affordance and then the ease with which you can then jump negative and how do we jump negative off of a proposition questions why what who where okay which are by the way look at one word easy dynamics why all right unbelievably tough to then answer but it's an unbelievably interesting on a proposition to add questions okay so the argument is this propositions quickly give rise to questions which then generates what's called a question answer dynamic which is the problem of justification okay justification is creating networks of legitimizing claims that coordinate the group and now this gives rise to then the evolution of justification systems the justification systems are the culture person plane of existence that evolves out of the primate layer and they're networked together through systems of symbolic justification okay and it makes a very clear prediction that says okay if that's the case then there's certain kinds of problems that come with propositions they're analytic problems like is it true or false alternative like what should we invest in as a group and personal like what should i share with the group relative to what i just keep to myself because the interesting thing about propositional knowledge is that it affords a clear window directly into the mental operations in the mind too in a way that action doesn't it's like well tell me what the hell you're thinking well that's a big goddamn problem why because you don't want everybody to know exactly what you're thinking all right so this is what's called the evolution in the unified theory this is the first key insight happens in 1996 prior to the evolution of the whole tree of knowledge it was oh my god the human ego is the mental organ

of justification okay the secondary narrator on top of the felt experiential self is tasked with generating what's justifiable in terms of what you feel how do you give reasons to maintain responsibility on the field and then how do you navigate your public persona relative to your private self and filter the dynamics and relationship because we do that you do that anytime you put a little lock on your diary you're filtering what people can gain access to okay we're constantly navigating what it is that people have access to relative to their justified in place and believe me now we're a clinician all the time okay you come with me the personas out here and their private cells in here why because they can't justify what the hell they actually think and feel to the world because they'll sacrifice social influence okay so the dynamic of justification goes through the skin and it creates the personal analytic and social problems which then actually gives rise to the structure of the ego which is hey you got to regulate your experiential self which by the way is freud's insight your it animal it and your ego and then your public self and then internalized introduction called your superego but then rogers and other people you know your public face out here irving goffman from a micro sociological analysis there are millions of different places that talk about the persona relative to the ego relative to and the justification problem then puts that very in clear evolutionary terms and what it does is by putting that puzzle of peace in you're like oh that's how a synchronized primate which was some gays and symbolic took off because the question answer dynamic exploded this new problem of justification and that basically hijacks the language acquisition device into or associates it with a self-conscious ego at justifier that then has to navigate the relationship to the public and the private okay and then groups get together and they build systems of justification which are what the belief values the coordinate of population it's like what do we believe in what do we value in how do we legitimize whether this guy's doing the right thing what is is an ought and how do we coordinate well the propositional networks of question answer dynamics build systems of justification and that's actually what we live in you know what we're doing right now is i'm in sharing with my system of okay so this thing takes off and then creates whoa this whole culture person space up here and evolves it evolves it right out of the primate and then that's what humans are they're both primates and persons okay and the person space is this reason giving responsibility so so now what that means is okay is that there's a fundamental division remember there's a division between living animals i mean living organisms and minded animals that's a joint point now we have a joint point between the relationship between the primate and the person which of course we all know there's no other goddamn primates paying attention to this talk right now my dog's wandering

around and he's heard this many different times he still doesn't get it so persons are able to get this you have to live at the level of justification all right why is that important to get back to the issue of psychology what is psychology okay many people wonder what do you mean by and one of the first things i realized with the subjective pro are we talking about animals in general which would be virtually all active inference stuff or are we actually talking about persons with reflective capacities in a social context that justifies and reasons about what they do some psychologists just think in terms of persons okay other psychologists the behaviors go all the way down hell the watson went all the way down to the goddamn amoebas without a nervous system okay so the the line of what you refer to in mental process is very different and people have different reference descartes gives us this reason justifying mind which i now call mind three so mind three and the culture person plane of existence is what this conversation's about mind two is the felt experience of being and mind one is the neurocognitive mind one a inside the nervous system mine one b moving around on the outside i got my things just died here i gotta switch over hold on ooh all once he connects back to the auditory input then i will continue then blue do you have anything first yeah i have some questions so uh let's just make sure he's back and then then go for it all right i'm now back sorry about that continue perfect perfect um so i i don't remember where i pulled a slide away from but but it's kind of relevant it's one of your slides but it's relevant to the talk about the distinction between self and other and these kind of complicated dynamics that are layered human human interaction and you know in this slide that i found it talks about the language of science as quantitative third person descriptive theoretical and public and also then the language of the subjective self which is described as qualitative first-person private um specific and value-laden and so i just was wondering if you have thought about second persons at all maybe you're familiar with the work of jay garfield just just the relationship between like like you don't really and jay garfield like there's the second person as soon as you realize that there's a you then then you realize that there's a me and so i think that that's quite fascinating and i just explained it for people that maybe haven't heard about it but but um is there a second person role in your framework totally so the slide i think you're referring to uh hold on i'll get one of these thanks [Music] let me reach into my exocortex yeah all of it's all cognitively offloaded into the niche um i think the slide you're referring to has this thing on it it does the iquad coin and then it has the tree of knowledge on the other side okay so all that that slide is saying is is that science is a particular kind of language system the thing that the primary thing i want to emphasize just in that slide is that science gives rise to a particular kind of language system and then i listed all of its

components and i would basically argue that the language of science and what do i mean by that it's like how do we build public knowledge systems that describe the generalizable world what do we mean by sort of really inter-subjective objectivity that then affords a scientific epistemology that then creates things like the periodic table of the elements okay um and it has a particular set of qualities the main thing that i want to emphasize in that slide is if you trace the evolution of science it basically gets more and more it evolves so that it eliminates the unique ideographic particular historical contingent subject at least the emphasis of that and over generalizes or emphasizes the generalizable nomothetic behavioral law vantage point so and then you can create a dialectic and and indeed the unified theory says hey there's a tree of knowledge that's your language of science and then the coin is actually the language of the subject okay so the coin then i got a picture of me on it and it's basically like oh this is my this represents it's actually in the shape of an h and you actually turn it to an i there's an i um and it basically stands for what's called the human identity function okay the human identity function refers to the unique particular subjective ideographic epistemological portal that is each one of us from our particular perspective that's what it represents okay which is not a scientific perspective uh so so so it captures science just basically factors that out it's like that's error from science spanish point okay it's our lives from our personal vantage point so now you that your question then is well do i engage in i mean what do we do about the second person that's a whole i totally in fact the garden is basically a second person wisdom collective so a second person is actually how do we create a we space okay a fundamental shared sense of being in a collective and what is a fundamentally about is actually wisdom is about how do we actually get collective intelligence together around an ethic around particular values that we can coordinate around and feel fulfilled and so totally it's just not position there but yeah we definitely have a second person perspective very nice i had a comment on science so um one distinction that i often think about is like science as an approach which could have happened anytime any place then there's the knowledge slash data the real specific data sets and then there's the real specific career and infrastructure so that just helps me separate the specific institutions that we know and love or don't from the data sets that are good or not from our approach which is timeless but then what's interesting about the slide it says the um is that the subjective self is the one that's specific in particular so i understand the sort of first and second person side but actually science as far as it being generalized and lawful the counter side is science is the framework by which we say well we don't prove anything we don't know that we're on the back of this animal actually we have to go look so science does particularize

through measurement and observation and the laws of science are probably a little bit overblown and that's the whole joke is that people outside of scientific training don't always have a sense of what scientists do it's not alchemy usually and then that is a disconnect that is also historical so it's still ground that's why we need this historical approach because we have to understand how does bacon's early writings play into some lay fallacies about what science is and then won't people just think differently if they change that belief beautiful and and now i'll point out there that yeah so another key issue here is the meaning of the word empirical okay um so the word empirical means observation systematic observation okay but early on the people didn't differentiate between oh i'm i empirically am observing this purple can but when science came along and you got the primary versus secondary quality dimension that emerged with people like hume and the whole structure you had the i experienced this is purple okay which then they were like well god that's a goddamn secondary quality which become qualia okay and then it's actually what we can really observe is the we can measure and create an inter-subjective objective intersection with and that becomes third person empiricism so the scientist is you have to get trained in procedures and it becomes a scientific observation if everybody trained in the procedure can see and interpret the data that you gather so science is not dependent upon a unique individual idiographic empirical perspective it's dependent upon a third person quantitative generalizable empirical perspective which is that equal to count as data you actually have to have transposed observers that are trained in interpreting and say oh yeah we all publicly agree that the way to interpret you guys look at these goddamn things coming off with a large hydrogen collider well we can get together i'll tell you what that is that's a muon that's wobbling in a weird way i'll look at that i don't know what goddamn muon is but actually all the experts together and they'll take the observation and say yep that's a muon and that's wobbling okay and that becomes an observation so it's a third-person data-driven observation that's available to those that have expertise in the procedures of interpretation great one way that that plays out i think um it makes me think about how in frequentism it's sort of like trial you know accept or reject kind of and misconceptions beyond that but with bayesian frameworks even though the rev base was back in the day the bayesian framework as it's now understood in the computational era helps us deal with probabilistic variables and variables we have uncertainty on and then also allows us which we use a ton in active is to have like an underlying process that isn't observed and then an emitted process that or an observable that is and having that as the fundamental that's signal processing general partially observable bayesian models so that's a very established

computational technique and so we can just take it as our unit that it's not just the measurements out there in the world nor is it an unmeasurable object you say we're always going to measure the dyad and that is the unit of scientific modeling because that's the tools we have in unified theory it's observer behavior relation cool um okay would you like i mean there's a lot there's a lot to go there are we gonna go somewhere else but anyway there's a foundational it basically takes science as observer behavior relation and then it it actually drops it into quantum mechanics and general relativity in a particular kind of way that aligns matrix mechanics with big bang it's weird but it basically goes oh yeah the fundamental unit in fact this is why this is an ipod coin it's i to the fourth which creates a unit circle and actually is a frequency of observer behavior relation is entangled in the coin so anyway just as i heard you're saying yep nope that's exactly what it is and really what science fundamentally is is observer behavioral frequency relations that's what lisa this argument is so i'd love to hear that okay uh short thought i hope it's already an nft but second thought is um what are the next steps even though we're not at the end of our conversation but what are the next steps for an individual or for the field like if someone's excited about this potential and maybe they're um haven't had a university education or they have like what does an individual do to onboard to know more to apply or what is the broader project that you're working on all right well that's uh yeah so uh let me see if i can uh so i gotta ship my frame a little bit here so let's see what we're talking about um so there's the there's a theoretical angle on this okay so the theor there's a theoretical angle which to me is that we're about ready to drop into a new theoretical enlightenment okay it's really exciting i really think that there is an opportunity to address the enlightenment gap and find a way to place the sciences in proper relation to our knowledge and do so with consilience that's super exciting from a theory perspective okay but it turns out and and so if that was going on that's beautiful and wonderful okay but i have to pause because that's not the only thing that's actually happening here at least from my vantage and that's the world is happening what do i mean by that i mean like the world is like in a very interesting spot right now okay and is happening at a very very um weird level okay why is this relevant for me well i'll just say okay you know i've been in the academy my whole life all right i've been i've i got i was at the university of pennsylvania worked for beck learned all that you know published its special issues blah blah trying to get the arc the institutional infrastructure to upgrade itself uh at least but i couldn't there was no i didn't i came at it from a two different position um and ultimately although i was inside the institution i couldn't get any leverage um and ultimately over the last four years things have happened so that i have

jettisoned myself outside the i'm still full professor and i do my job okay but actually a wave has happened inside the institution that i feel qualitatively more alienated from over the last four years and that's for personal reasons okay and retrospective reasons in relationship to my difficulty trying to get traction and global reasons and zeitgeist reasons so that now i hang out with people in that are futurists okay so my my i used to spend you know in traditional blah blah and i still do but i actually spend all my time talking with people that are in the midst of a zeitgeist like for example i'm involved in like metamodernism okay metamodernism is an emerging sensibility okay that is basically the simplest way to think about it is the thing that comes after post-modernism that looks back and resolves the modernist versus postmodern modern thesis post-modern antithesis meta-modern synthesis okay and there's a emerging sensibility in europe about this which emerges over the last decade okay and weird happens with that so like the an ethos is sort of like it's a it's seeking a post-post-modern grand meta narrative so post-modernism says there is no grandma narrative they're always dangerous and this one comes after and says i wrote in my book in 2011. and i was like hey the unified there is a post post-modern grand better narrative and then i turned this into a garden and i don't know why the hell i did that you know that's weird okay but the issue is is that when i then saw the artistic sensibility of metamodernism it's called sincere irony okay sincere in the sense it's authentic and true irony in a sense it's a critique and twisted well that's what that is this is unbelievably sincerity and it's a cartoon and it's twisted and it's both science and religion and it's a joke and it's serious and it's all that i had no idea that the meta-modern sensibility had a sincere ironic flavor i was called to do this okay oh i'm connecting to people now the experience that i have is like the the i don't know if you saw the close encounters are the third kind all right close encounters occurred kind of like 10 000 people in the world sort of wake up and they get this image of devil's mountain all right and he gets his callings crazy calling like oh my god oh my god and then everybody migrates to it and then blow the whole that's where the aliens show up okay well i'm telling you i feel like there's a calling that's happening for in the process of a particular serious evolution we're in a serious evolutionary flux okay and and the tree of knowledge actually clarifies exactly why we would be remember each one of those life mind culture what does it say each one of those is an information processing and communication network system that yolks stuff together at a higher order huh okay so cells did that and then animals did that and then language did that is it any possible that there could be an information processing network communication system that is emerging at a higher order well you actually do that you look back and you're like oh my god that's a goddamn 20th

century we laid down the internet and we built artificial intelligence systems that are intersecting with it so from a tree of knowledge perspective we're actually in the midst of the fifth joint point on this century this century is the fifth joint brain when you're traveling through the culture person plane of existence into the potential digital it's a whole new potential metaverse of complexification okay if you see the world through that particular ends then you're like whoa now we can actually track the evolution of complexification across all these different frequencies levels with at least intuitive comprehensibility and you can then see well we're actually in the midst of a flux point as we have built all of these different technologies that transcending the old infrastructures for institutional control and regulation sense making and we're all going insane because this is opening up all sorts of possibilities but it's also threatening the fundamental foundation of how we operate and then the question is can we upgrade our wisdom to navigate the evolution of technology and the planet and do so in a way that stabilizes or is there going to be too much goddamn chaos and flux the system's going to be thinly stacked everybody gets defensive and insane because of the new thing and the goddamn thing plummets blue buy sell yeah intense i mean i don't know i'm lost in the in the artificial intelligence and cognitive offloading and niche modification i think i i wonder about what you feel like the role is since you brought up the internet um and artificial intelligence like what do you feel like the role is of um niche modification in the transmission of culture or do you feel that culture is something that has to then be transmitted directly like is that a person to person a learned behavior or can we modify the niche and transmit it in that way well so let's to be clear about some definitions so in the true knowledge it's capital c culture okay then there's lower c culture but capital c culture is the technical term for the systems of the propositional networks at various levels that legitimize is an ought in inside groups or outside of them so if you're inside a culture you have a propositional network of legitimizing is an art that can be coordinated with yourself at the ego level diatically or move on up the stack so if you take act inference lab and you create a culture on it it will be networked around a system of justification that legitimizes what it is that you're doing in a particular niche and that creates the act inference culture capital city propositional network there's little sea culture which is basically the behavioral investment patterns other animals have little seed culture okay that then afford the capacity to adjust into niches and then feed back on them you know animals learn how to work on washed potatoes no monkeys will do that all right um and you can then and of course we're unbelievably flexible in our behavioral repertoires all right so now the fundamental issue is that we are then uh from a macroscopic

level what are the sys what are the large-scale systems of justification that are operating to coordinate is an awe and the answer is not much because of the chaos in the globe the globe has afforded us to get together but our knowledge systems are not up to the task of coordinating we know where to learn it this is the metaverse i mean what the hell is this thing okay and so we're trying to we're groping to build justification systems that can effectively hold and cultivate all the niches that actually then need to grow into that afford individual variation and new growth and blah blah and at the same time do so in a way that you know well makes this transition or at least affords the emergence of this whole new way of relating the digital okay that can collapse space and time and have us talk to each other at the exact same time which is totally weird right um i guess my basic point is there's multi-level operations all the time okay uh that are down more trans-determinative meaning there's everything is sort of happening across the synaptic diachromatic stack okay well i'm not a rigid determinist by any stretch um but anyway then our i see us as having a moment of conscious evolution like the issue fundamentally for us um at a broad level is to wake up to the situation we find ourselves in okay and upgrade our operating system to be up to the task upgrading our system to gain knowledge and socialization so we can be effectively oriented toward wisdom that's what the fundamental garden is basically there is a way um that we can then and then if we solve some of the old problems that have prevented this upgrade from happening so that we can see the path to the upgrade then a map of cognitive architectural map from the sort of the intellectual stream will happen if that's in good contact with people that are realizing that the system is sick right now and want a community upgrade then the intellectual solution can feed the community upgrade they can metabolize it and get it to normal people living their lives and we can change the contingency infrastructure nation state capital labor relations resource control stuff and hopefully then evolve so to me the issue is very flux very bottom up top down lots of different things will influence at the same time my goal is really there's a meaning in mental health crisis that's one aspect of the dynamic we face that's what the unifi you talk fundamentally then it's oriented towards so my life mission is we can do a lot better in meaning making and mental health than we currently are there are ways to understand knowledge and wisdom that can fundamentally shift and rotate the way we think about ourselves so we can in fact i'm searching for a coherent naturalistic ontology that can revitalize the human soul and spirit in the 21st century that's essentially what my the mission is and utah affords a particular sense-making structure to do that and if everyone's operating in that kind of sense-making structure from both the bottom and the top the whole coordinated

system of justification would be different and the consequences of many many things would be different thanks for sharing that just to connect it to two or a few active ideas policy selection which is core based upon the partitioning but always the joint consideration of epistemic value knowledge oriented value or intrinsic motivation and pragmatic value utility external motivation so we need to learn more you know we got to take a few steps back to go the easier way to the mountaintop but also we need to clear the clouds to see where we could go and communicate how we could go so that's why planning as inference is a useful framework because it's not enough to just have preferences preferences are a core part of our models we want to be like warm and secure at each level of the maslow's hierarchy but the other side that in activism kind of slapped back into the world we have to have a plan and it has to be pragmatic just beautiful very similar it's funny because drawing more from the cybernetics dynamic systems and complex systems modeling side the experience that's the part that's oh that's unsupported or you'll never be able to make those claims or that's you know that's the pl gangplank versus the kernel which is seen as to be the math but then the real relational insight shows that it's truly a bi-directional freeway like everyone's going home or away because you really can wrap a lot of the technical parts within a broader picture that isn't formal amen brother so just a quick question uh related to both the mental health crisis and to culture with the capital c um which i don't know is in that kind of definition like something i'm equating to like a moral system or a value system is that is that appropriate yeah oh sure yeah it's a it's a belief value network of propositions that afford a group to be able to state explicitly what is an awe and negotiate from there so what do you think um or do you think that there is a relationship between like the fact that you know in western society at least in the states like many of us are just spiritually bankrupt and so like do you think that there is a connection is that a possible fundamental reason for the mental health crisis that that we're facing now do you think that those are linked exactly no directly 100 yeah no no we we we've we're operating off of a very sick system of justification and and um it's basically my friend zack stein puts it this way um or at least i'll draw from him and i'll put it in my own basic context so what happens with the modernity which is basically enlightenment capital labor relations uh democrat and then the evolution and what modernity does to the traditional knowledge systems and the oral indigenous knowledge systems prior to that okay is it basically shifts the structure into a skill development um self-interested control model okay basically so now the values of justification that evolve off of the modernity are not spiritual insolent but ultimately reductive physical self-interested control mechanistic kinds of instrumental ways of being

okay which is an important way of being no doubt um but it's a doing problem okay which is different than eric from we talk about as a being problem all right let me give you an example which i stumbled into through the unified theory to explain this okay so the fourth branch back there is this thing called the influence matrix all right and what it does is it basically combines attachment theory and the interpersonal circumplex uh to give a model of the process dimensions that we track in our dyadic relational world okay meaning the self-other world and then we wander around with internal working models in relationship to it so the the core line in the influence matrix i originally called uh the social influence line that's why it's called the influence matrix and it said that we had high social influence or low social influence okay and influence here was defined in pretty instrumental terms the instrumental terms were the capacity to influence others in accordance with your interests or the reverse the absence of that so there's a resource in conception of influence and it's the capacity and that's becomes the fundament and from a pure kind of a standard evolutionary mechanistic modernist kind of view this capacity to influence the world is where rubber meets the road okay but over time that thing evolved and it became the relational value and social influence line whereby what we're actually tracking is both our capacity to move other people instrumentally and the extent to which we're known and valued by important others okay and that is our heart our heart is both hey can we actually move people around and be am i genuinely known and valued by important others for my essence okay and what this emerges in my clinical work okay is a i work some with some people who had no social influence and no relational value homeless people who are brutalized by all aspects of life and they're just miserable but i then met with a number of people of course this will come as no surprise who actually had high social influence who were completely empty they manipulated and controlled other people but they didn't they had they stacked themselves up with high resumes but then nobody liked them they could be bosses and control people and had money okay so they had instrumental influence but and of course then if the instrument just changes you now have to spend all your time especially if you're going to do it through control to gain influence through power and now that's a very vulnerable spot you lose access to control and you're jettisoned and then there are other people even who were actually valued by other people but they put on a face their persona is what they felt was value they weren't known okay and these people also felt very anxious because they could be discovered for who they were the core heart placement of security is to have social influence and be known in value for your essence okay hunter-gatherers get this we because of our instrumentality in the way we decide to control then we

create economic labor relations we basically turn everything into an influence game and we lose relational value okay we don't know that actually that's really what i mean uh you know john watson basically decided that the reason kids wanted hugs was because it was paired with food i mean that's how insane his model of humanity was and by the way two of his three kids killed themselves that's john watson behaviors okay so our idea of self-interested mechanistic control and applying the clockwork way in which power force mass and acceleration actually work as if really that's the underlying causation of our true essence actually kills our soul because it turns it into instrumentality and fails to attend to the genuine beingness and the value of that we have to revitalize that and we know it when we feel it okay but our justification systems meaning our academic hard-nosed tough-minded thing was basically well when it really comes down to it you're basically just a bunch of molecules i mean that was the message okay and we need to change that message people digested that message people lost the clarity about what a human soul and spirit is from a naturalistic perspective we can revitalize that get clear about it point them to it and if we build systems of justification that then organize institutions like education politics national relations et cetera the whole goddamn change equation changes at least according to this so i it just brings me to another question that i had unless daniel you have something that you wanna about blue with the last question and then we'll have a closing thought so thanks a lot greg this is really a cool dialogue so blue last question and then we'll each give a last thought so um just one final question is speaking of you know relational interactions and you know if the relational nature isn't food and it isn't like we're not billiard balls we're existing in you know a field essentially like what is the relational language like how can we best relate to one another relate to ourselves like how do we speak that dialogue beautiful um well so i built this thing called map of mind to just defer metaphysics of the mental domains and there are five domains that are very obvious okay so one is mind1b that's okay mind1a that's what my nervous system is doing okay mine two that's the i can see but you can't mine three a is what i talk to myself okay i was just doing that mind3b is now me talking out here okay so and those are all easy any behavioral no matter how empirical they are you can now have a map that says oh these are actually the meanings of referent of mind and notice there are five of them and we use the word mind and actually there are five different reference here we need to get very clear because they're different epistemological and ontological reference points okay here's one that um gets right to your point okay i think and i think michael tomasello's work and i think anybody that really falls in love and feels is mine to be the inter-subjective non-verbal

intersubjective space of tracking one another and participating in the attachment relational dance of being it's seeing the other person through the eyes tracking where their eyes are and essentially seeing their heart their felt experience of being and being part of that and then being in dance with that and if anybody's ever fallen in love with somebody okay and then you fall into that and then all of a sudden you are that field that felt experience of being you know joins together in a particular way and i actually think there will we can develop you know genuine metaphysical claims about mind to be space which is an implicit inner subjective space i think tonicello's work on shared attention and intention i think many different things domain in relationship to that and it basically is a relation a pre-verbal primate relational field that we are actually really attuned to and i think that that informs us a way to think about how we sync up with each other relationally i think we should shine a light on that because i think we're relationally deprived and i think that many things like social media basically break that completely relative to what it is designed to track and so you're seeing a mental health crisis in adolescence they don't have any mind to be connection everything is texting and likes and you know jolting and everything else and the whole sense of who i am and who you are i mean you go back to hunter gatherers you lived in a family forever and they saw all of this it was real world action i don't know and i think that's not what the heart needs and some that we got to get we got to figure out a way uh to cultivate that or else we're going to be empty and so yeah let's shine a light on that mind to be relational world it's either going to be 2b or not to be brilliant thank you touche or whatever really at least love it i'll give a last comment or blue do you want to go first and then to greg for the final um i think that this was you know especially the last remark was just very beautiful and very poignant um so thanks greg for coming on and it's been a pleasure see you later but really any time that first off if anyone wants to get involved in active lab hopefully even in the future if you're watching this we're still going to be there to help scaffold and um yeah greg if you or any colleagues or anyone else ever wants to get involved or do another discussion or anything else this was really a great convo beautiful well thank you so much it's a really honor to be here i really enjoyed it um i'm interested in us cultivating wisdom energy that's that's one way of summarizing what the project is here and uh for folks that you asked like what can we do ultimately yes look out have some concern look around at your local garden and try to cultivate wisdom energy in your local garden with your heart yourself and the other people that you love in the world and we do that enough when we might be all right and we have the ontology working group in the lab for when that's not all right right well this and i and as a theorist that's absolutely essential

the ontology the failure of ontology enlightenment gap is part of why we're in the mess if we
can get this ontology right we can get on the right path 100 thank you greg talk to you later see
you blue thank you brilliant take care okay